```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Colored Scatterplots

```
iris = pd.read_csv('iris.csv')
iris.sample(5)
```

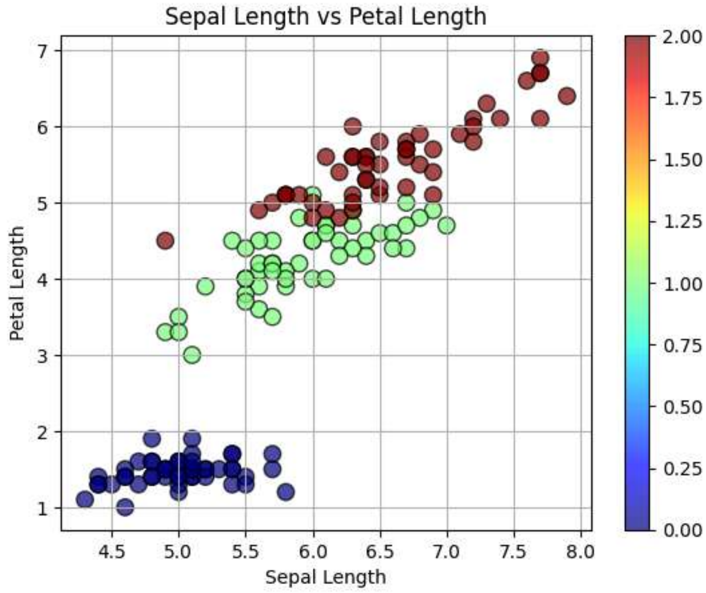
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
10	11	5.4	3.7	1.5	0.2	Iris-setosa
113	114	5.7	2.5	5.0	2.0	Iris-virginica
59	60	5.2	2.7	3.9	1.4	Iris-versicolor
79	80	5.7	2.6	3.5	1.0	Iris-versicolor
80	81	5.5	2.4	3.8	1.1	Iris-versicolor

```
iris['Species'] = iris['Species'].replace({'Iris-setosa':0,'Iris-versicolor':1,'Iris-virginica':2})
iris.sample(5)
```

```
/tmp/ipykernel_1166/572492303.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To ret
  iris['Species'] = iris['Species'].replace({'Iris-setosa':0,'Iris-versicolor':1,'Iris-virginica':2})
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
9	10	4.9	3.1	1.5	0.1	0
48	49	5.3	3.7	1.5	0.2	0
115	116	6.4	3.2	5.3	2.3	2
140	141	6.7	3.1	5.6	2.4	2
69	70	5.6	2.5	3.9	1.1	1

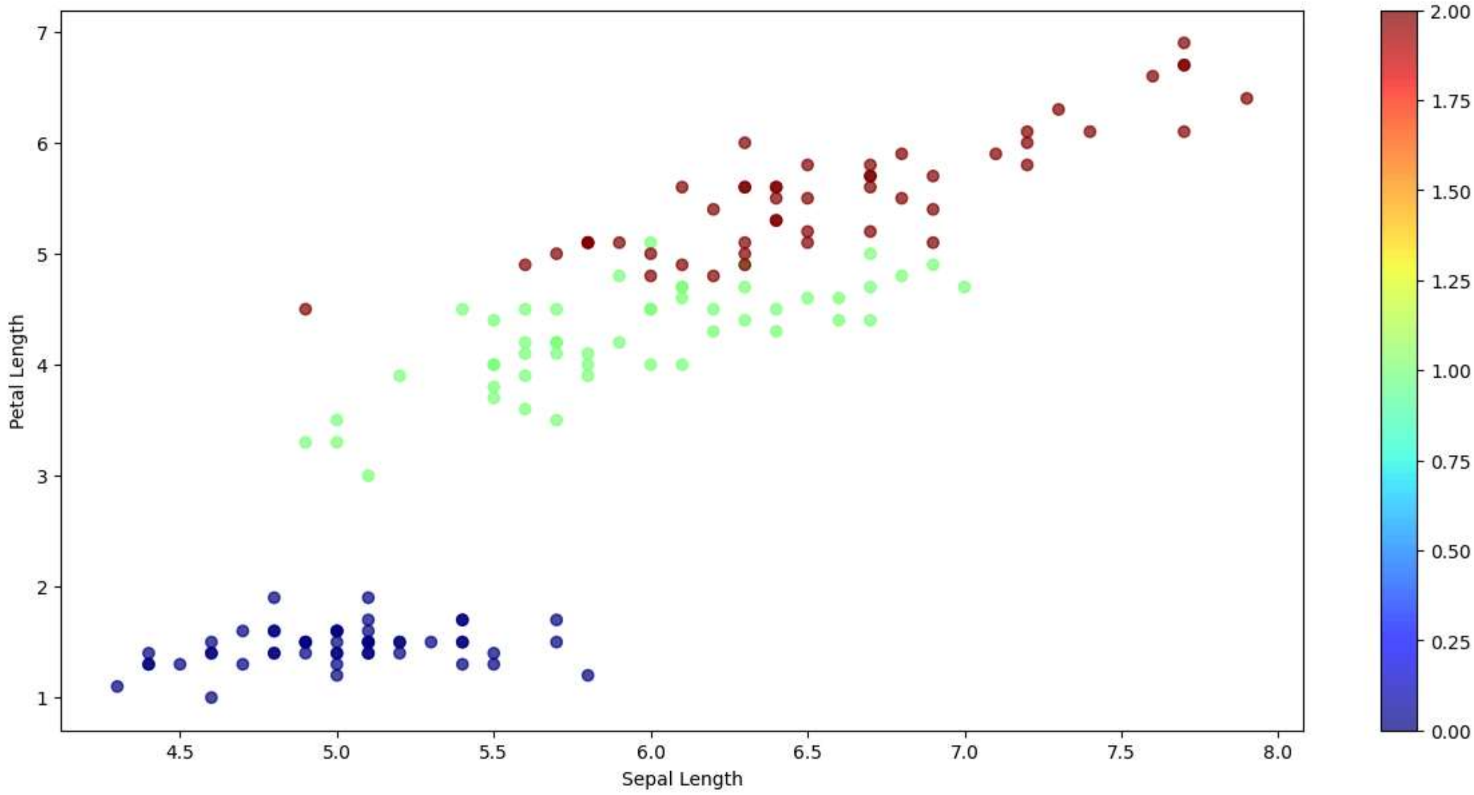
```
# plt.scatter(iris['SepalLengthCm'],iris['PetalLengthCm'],c=iris['Species'],cmap='jet',alpha=0.7)
# plt.xlabel('Sepal Length')
# plt.ylabel('Petal Length')
# plt.colorbar()
import matplotlib.pyplot as plt

plt.scatter(
    iris['SepalLengthCm'],      # x-axis values
    iris['PetalLengthCm'],      # y-axis values
    c=iris['Species'],          # color based on species
    cmap='jet',                 # colormap
    alpha=0.7,                  # transparency
    s=80,                       # size of points
    marker='o',                 # marker style
    edgecolors='black',         # border color of points
    linewidths=1                # border thickness
)

plt.xlabel('Sepal Length')      # x-axis label
plt.ylabel('Petal Length')      # y-axis label

plt.title('Sepal Length vs Petal Length')   # graph title

plt.grid(True)                  # show grid

plt.colorbar()                  # show color scale

plt.show()                      # display plot
```



Plot size

```
plt.figure(figsize=(15,7))

plt.scatter(iris['SepalLengthCm'],iris['PetalLengthCm'],c=iris['Species'],cmap='jet',alpha=0.7)
plt.xlabel('Sepal Length')
plt.ylabel('Petal Length')
plt.colorbar()
```

```
<matplotlib.colorbar.Colorbar at 0x7f59e8ff6690>
```



Annotations

```
batters = pd.read_csv('batter.csv')
```

```
batters.shape
```

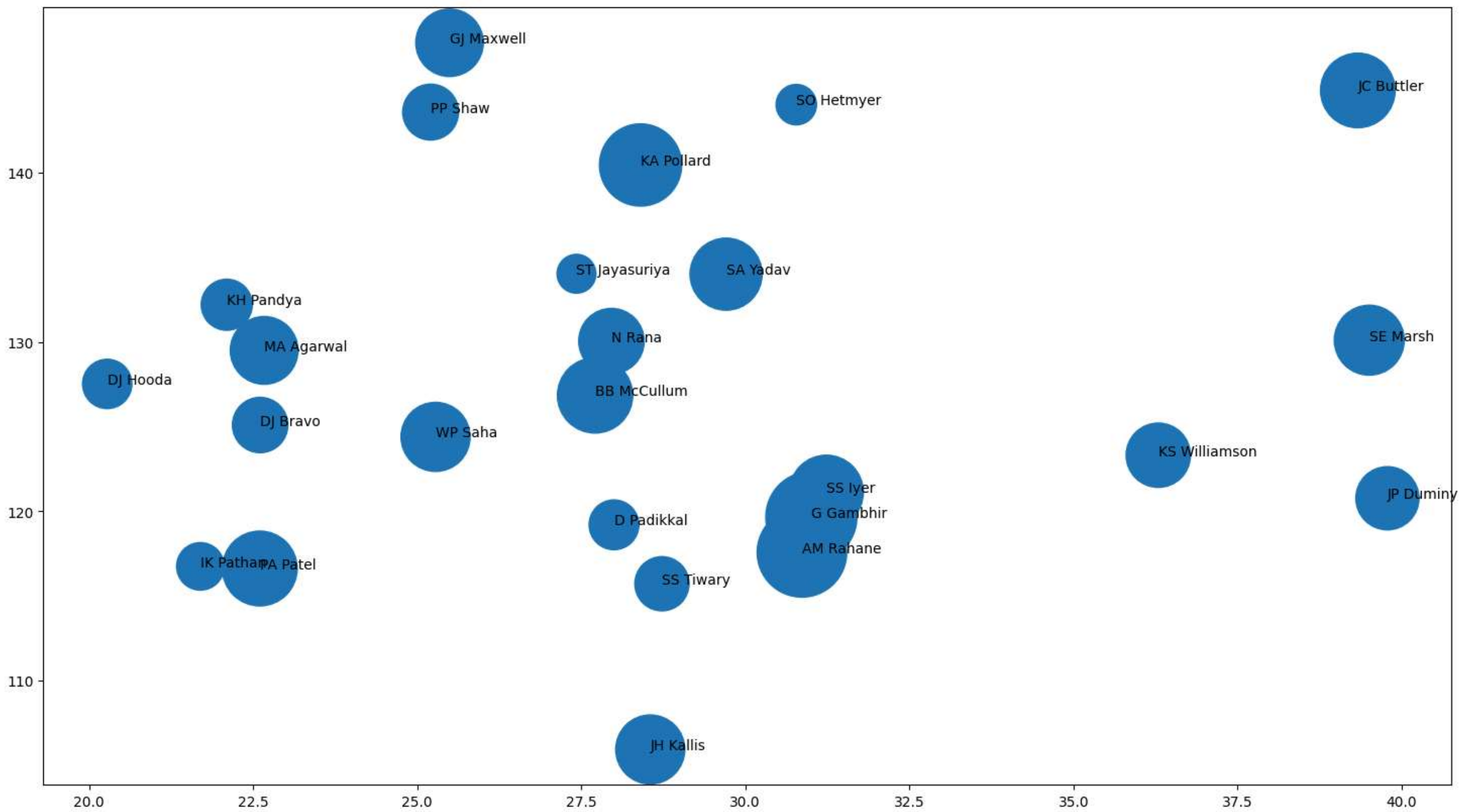
```
(605, 4)
```

```
sample_df = batters.head(100).sample(25,random_state=5)
```

```
sample_df.head()
```

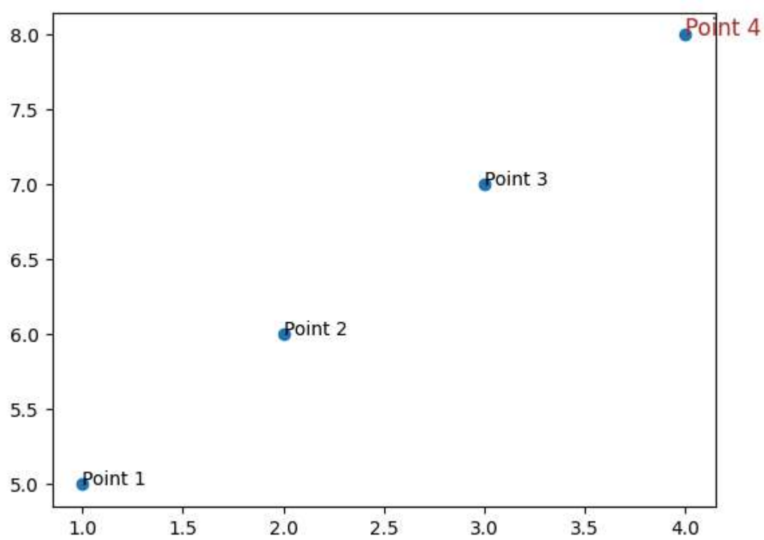
	batter	runs	avg	strike_rate
66	KH Pandya	1326	22.100000	132.203390
32	SE Marsh	2489	39.507937	130.109775
46	JP Duminy	2029	39.784314	120.773810
28	SA Yadav	2644	29.707865	134.009123
74	IK Pathan	1150	21.698113	116.751269

```
plt.figure(figsize=(18,10))
plt.scatter(sample_df['avg'],sample_df['strike_rate'],s=sample_df['runs'])

for i in range(sample_df.shape[0]):
  plt.text(sample_df['avg'].values[i],sample_df['strike_rate'].values[i],sample_df['batter'].values[i])
```



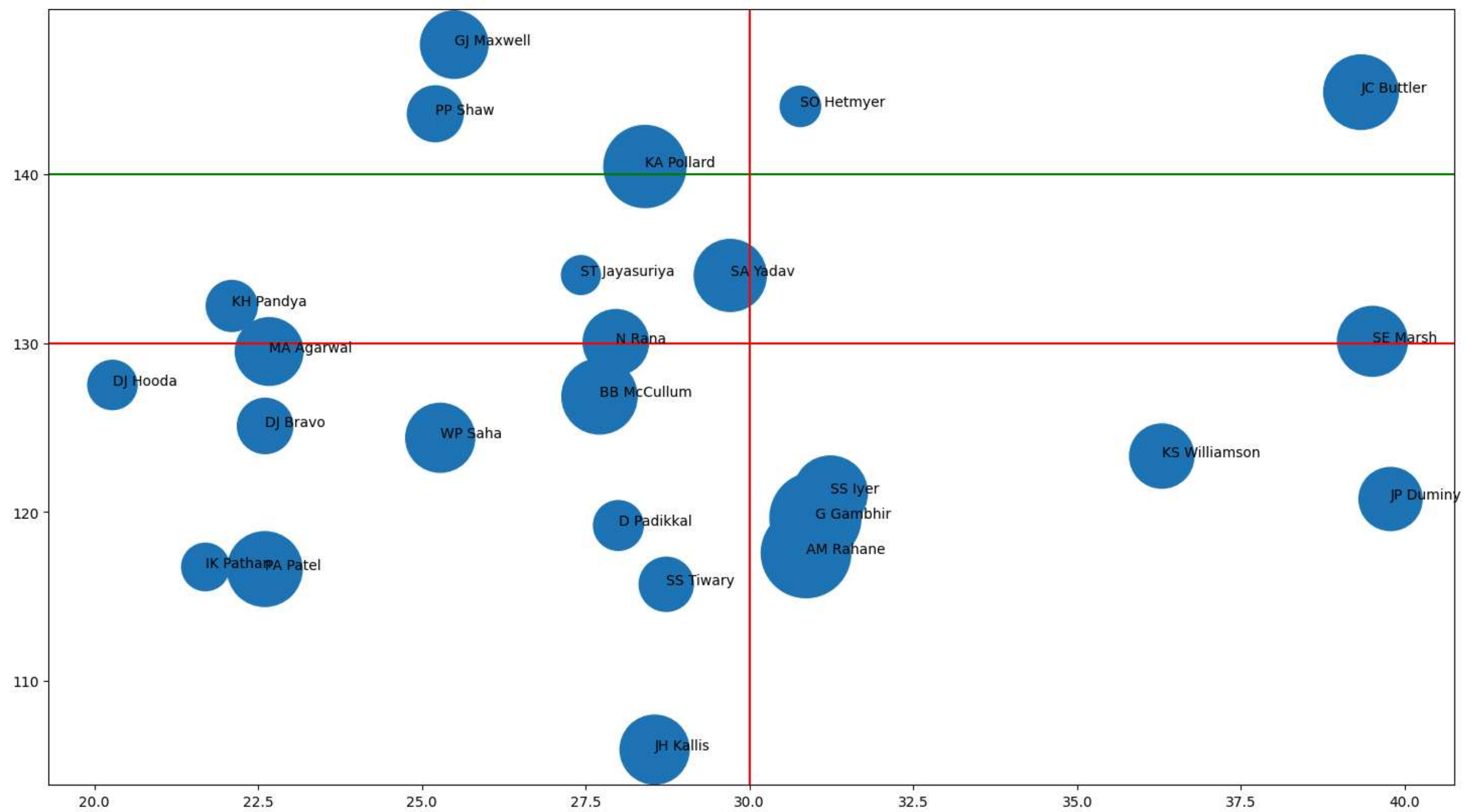

```
x = [1,2,3,4]
y = [5,6,7,8]

plt.scatter(x,y)
plt.text(1,5,'Point 1')
plt.text(2,6,'Point 2')
plt.text(3,7,'Point 3')
plt.text(4,8,'Point 4',fontdict={'size':12,'color':'brown'})
```

```
Text(4, 8, 'Point 4')
```



Horizontal and Vertical lines

```
plt.figure(figsize=(18,10))
plt.scatter(sample_df['avg'],sample_df['strike_rate'],s=sample_df['runs'])

plt.axhline(130,color='red')
plt.axhline(140,color='green')
plt.axvline(30,color='red')

for i in range(sample_df.shape[0]):
  plt.text(sample_df['avg'].values[i],sample_df['strike_rate'].values[i],sample_df['batter'].values[i])
```

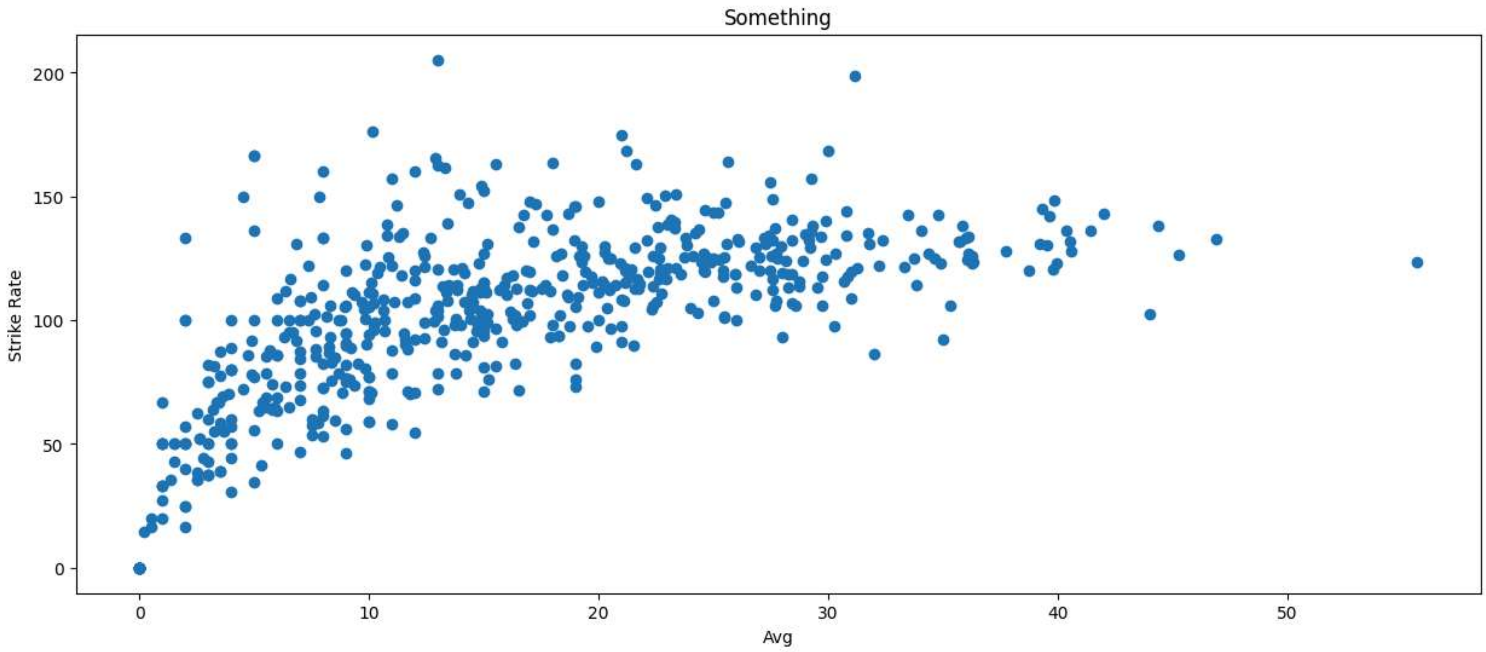


Subplots

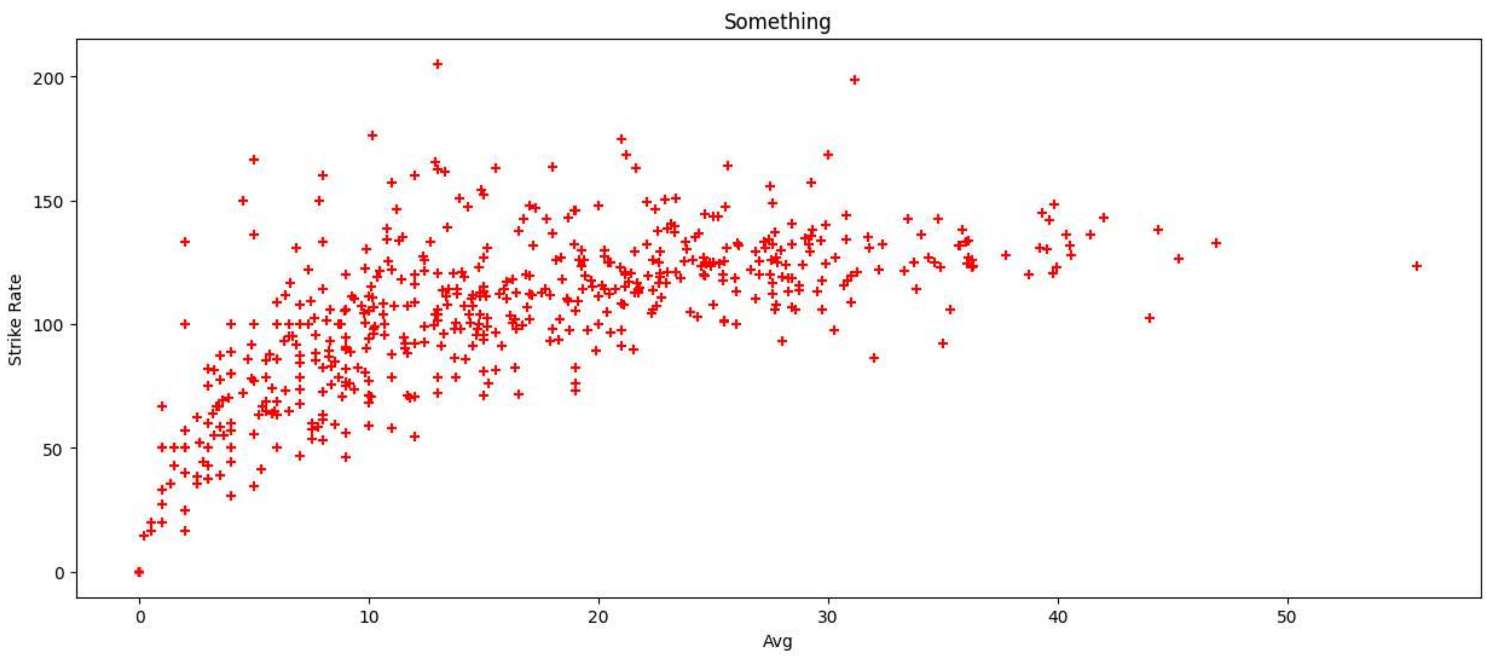
```
# A diff way to plot graphs
batters.head()
```

	batter	runs	avg	strike_rate
0	V Kohli	6634	36.251366	125.977972
1	S Dhawan	6244	34.882682	122.840842
2	DA Warner	5883	41.429577	136.401577
3	RG Sharma	5881	30.314433	126.964594
4	SK Raina	5536	32.374269	132.535312

```
plt.figure(figsize=(15,6))
plt.scatter(batters['avg'],batters['strike_rate'])
plt.title('Something')
plt.xlabel('Avg')
plt.ylabel('Strike Rate')

plt.show()
```




```
fig,ax = plt.subplots(figsize=(15,6))

ax.scatter(batters['avg'],batters['strike_rate'],color='red',marker='+')
ax.set_title('Something')
ax.set_xlabel('Avg')
ax.set_ylabel('Strike Rate')

fig.show()
```



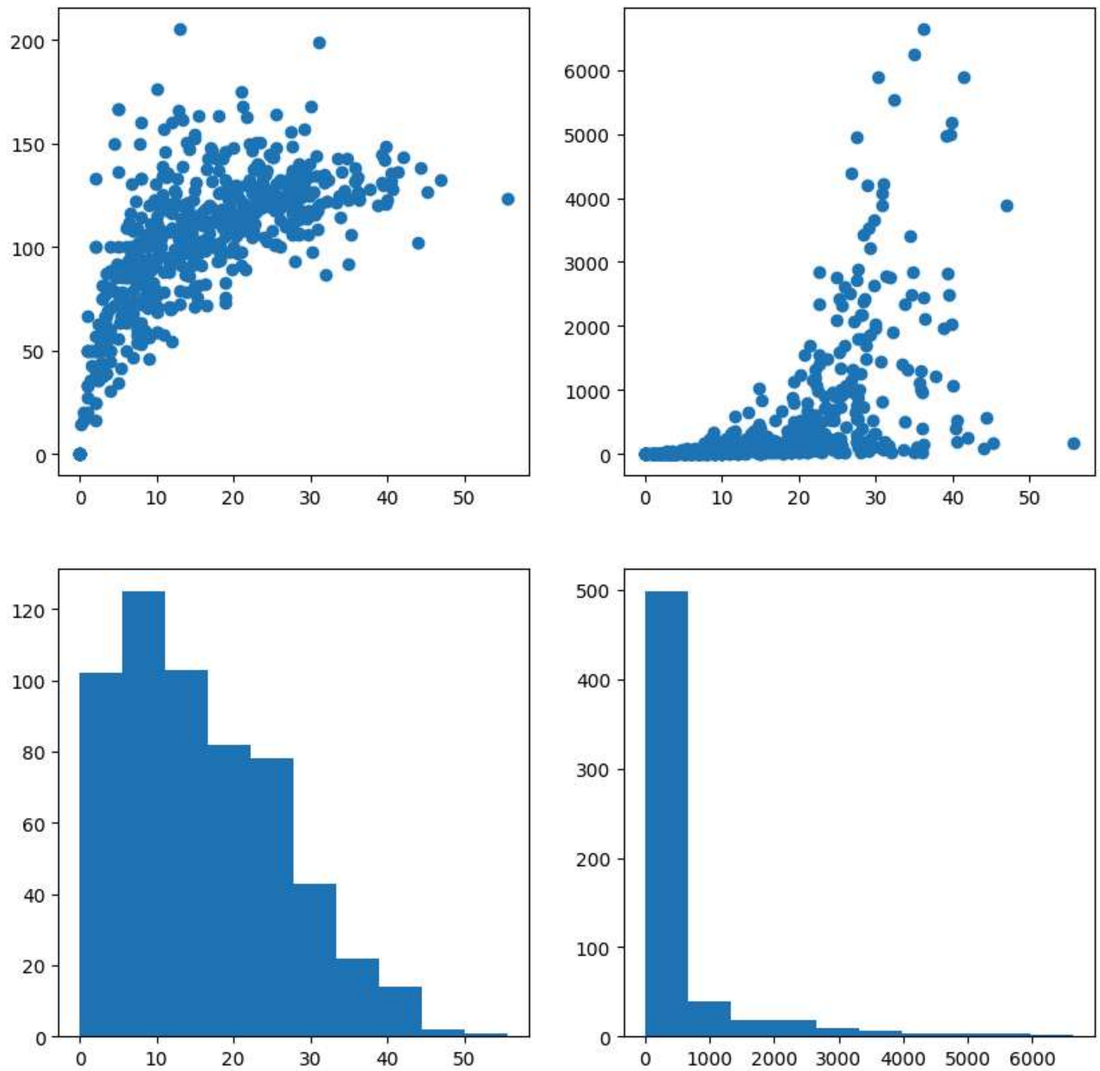

```
# batter dataset
```

```
fig, ax = plt.subplots(nrows=2,ncols=1,sharex=True,figsize=(10,6))

ax[0].scatter(batters['avg'],batters['strike_rate'],color='red')
ax[1].scatter(batters['avg'],batters['runs'])

ax[0].set_title('Avg Vs Strike Rate')
ax[0].set_ylabel('Strike Rate')


ax[1].set_title('Avg Vs Runs')
ax[1].set_ylabel('Runs')
ax[1].set_xlabel('Avg')
```

```
Text(0.5, 0, 'Avg')
```


```
fig, ax = plt.subplots(nrows=2,ncols=2,figsize=(10,10))

ax[0,0].scatter(batters['avg'],batters['strike_rate'])
ax[0,1].scatter(batters['avg'],batters['runs'])
ax[1,0].hist(batters['avg'])
ax[1,1].hist(batters['runs'])
```

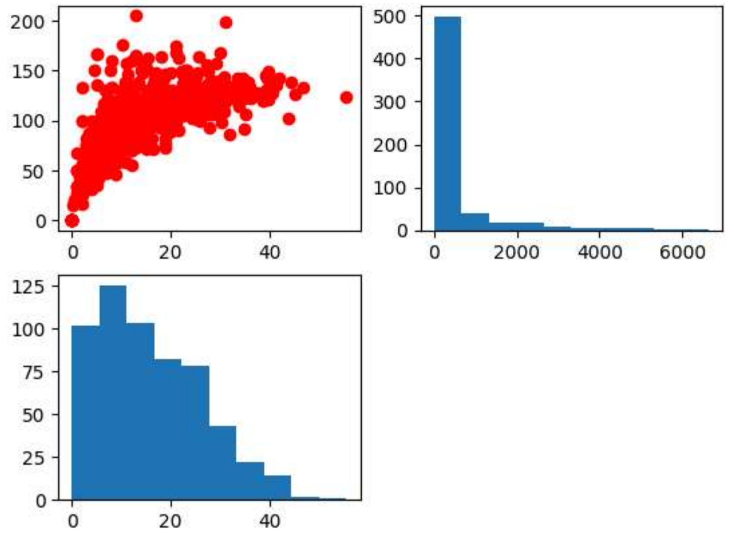
```
(array([499.,  40.,  19.,  19.,   9.,   6.,   4.,   4.,   3.,   2.]),
 array([   0. ,  663.4, 1326.8, 1990.2, 2653.6, 3317. , 3980.4, 4643.8,
        5307.2, 5970.6, 6634. ]),
 <BarContainer object of 10 artists>)
```



```
fig = plt.figure()

ax1 = fig.add_subplot(2,2,1)
ax1.scatter(batters['avg'],batters['strike_rate'],color='red')

ax2 = fig.add_subplot(2,2,2)
```

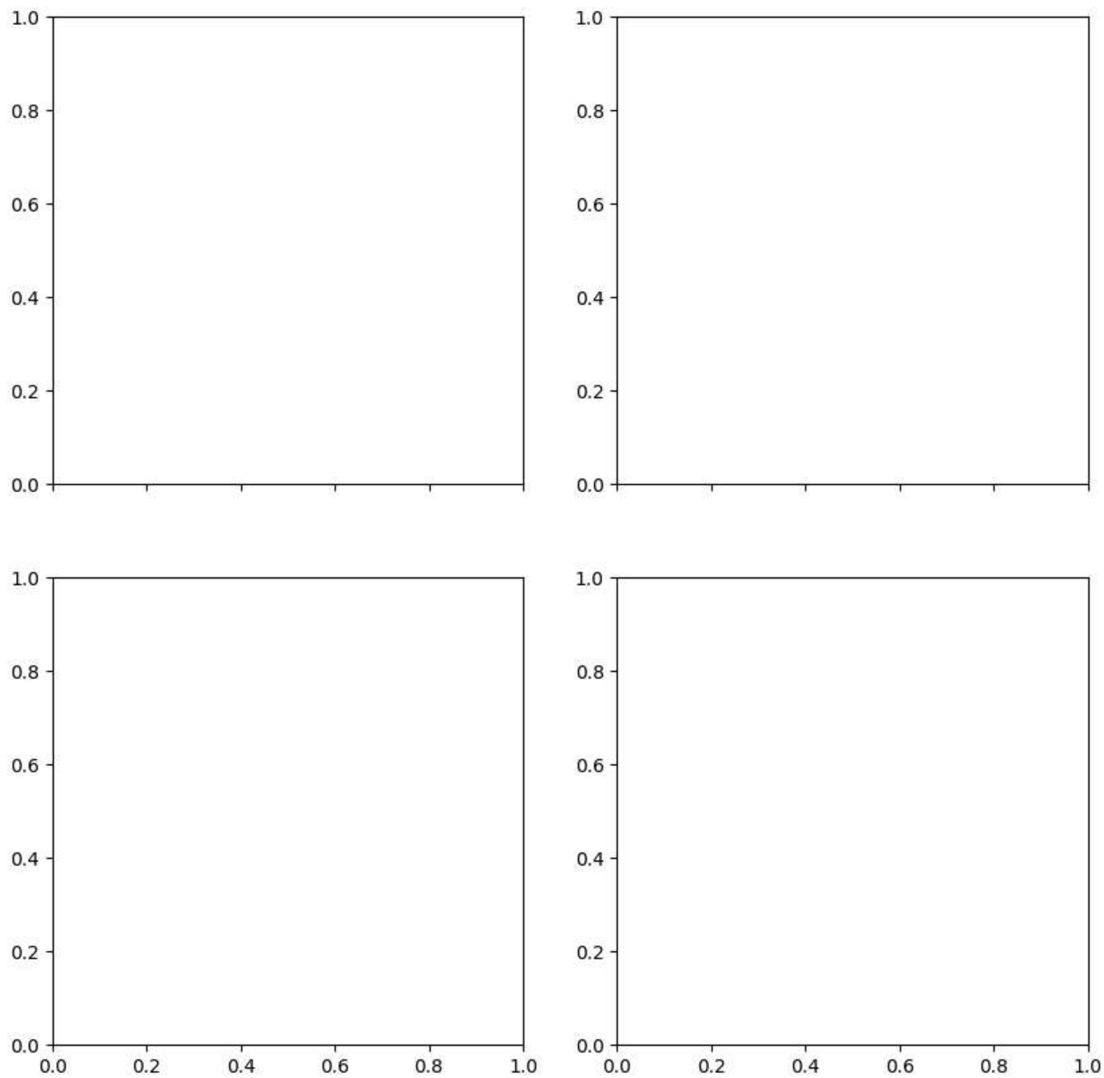
```
ax2.hist(batters['runs'])

ax3 = fig.add_subplot(2,2,3)
ax3.hist(batters['avg'])
```

```
(array([102., 125., 103.,  82.,  78.,  43.,  22.,  14.,   2.,   1.]),
 array([ 0.        ,  5.56666667, 11.13333333, 16.7       , 22.26666667,
        27.83333333, 33.4       , 38.96666667, 44.53333333, 50.1       ,
        55.66666667]),
 <BarContainer object of 10 artists>)
```




```
fig, ax = plt.subplots(nrows=2,ncols=2,sharex=True,figsize=(10,10))

ax[1,1]
```

```
<Axes: >
```

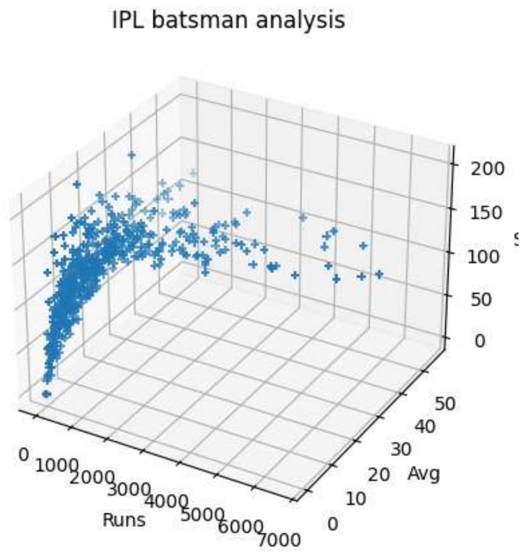


3D Scatter Plots

```
batters

fig = plt.figure()

ax = plt.subplot(projection='3d')

ax.scatter3D(batters['runs'],batters['avg'],batters['strike_rate'],marker='+')
```

```
ax.set_title('IPL batsman analysis')

ax.set_xlabel('Runs')
ax.set_ylabel('Avg')
ax.set_zlabel('SR')
```

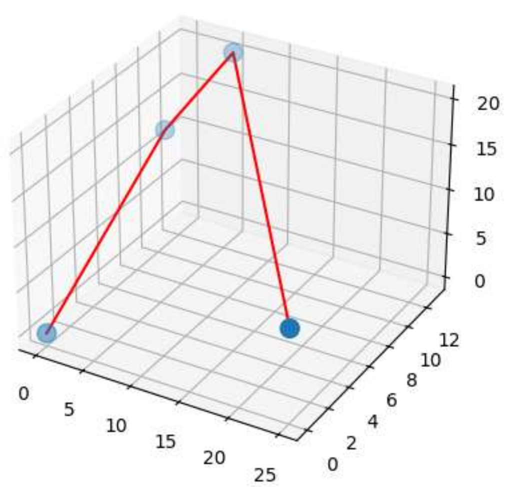
```
Text(0.5, 0, 'SR')
```



3D Line Plot

```
x = [0,1,5,25]
y = [0,10,13,0]
z = [0,13,20,9]

fig = plt.figure()

ax = plt.subplot(projection='3d')

ax.scatter3D(x,y,z,s=[100,100,100,100])
ax.plot3D(x,y,z,color='red')
```

```
[<mpl_toolkits.mplot3d.art3d.Line3D at 0x7f59e2326960>]
```



3D Surface Plots

```
x = np.linspace(-10,10,100)
y = np.linspace(-10,10,100)
```

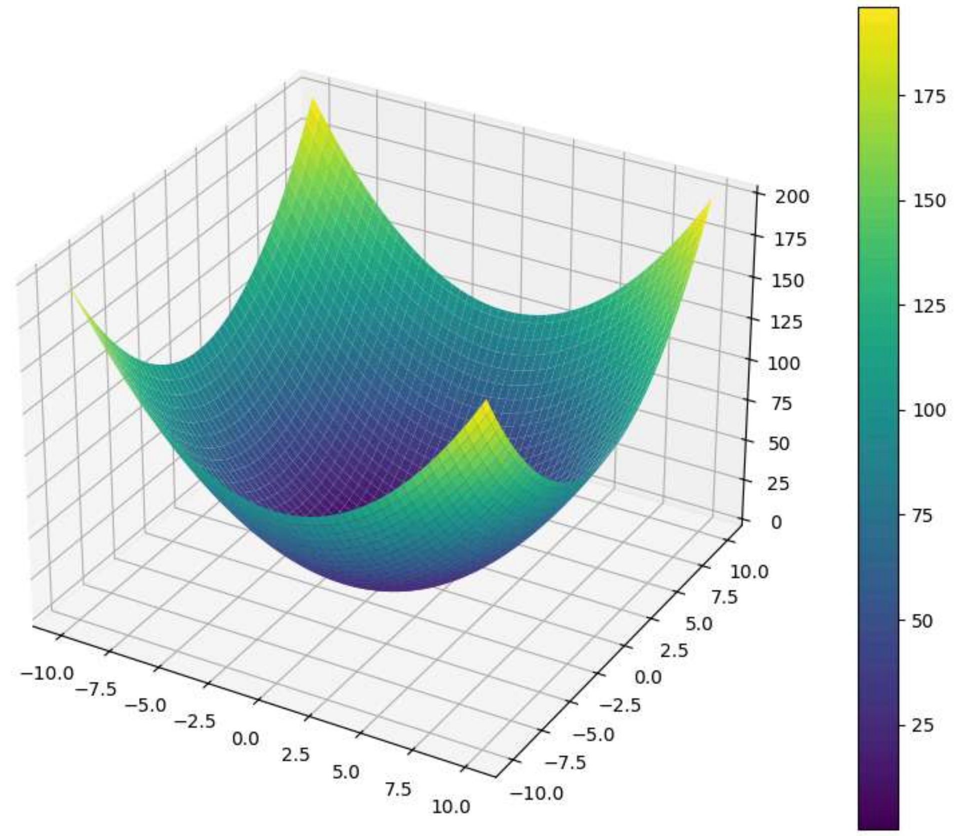
```
xx, yy = np.meshgrid(x,y)
```

```
z = xx**2 + yy**2
z.shape
```

```
(100, 100)
```

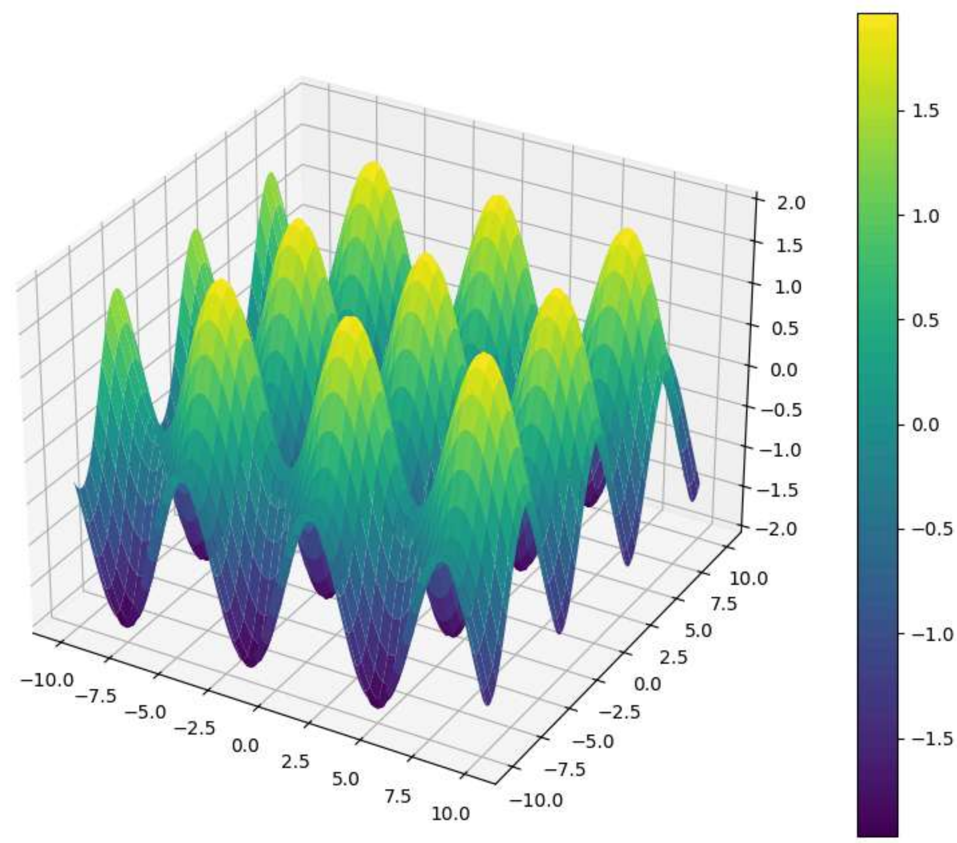
```
fig = plt.figure(figsize=(12,8))

ax = plt.subplot(projection='3d')

p = ax.plot_surface(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
<matplotlib.colorbar.Colorbar at 0x7f59e21cb4d0>
```



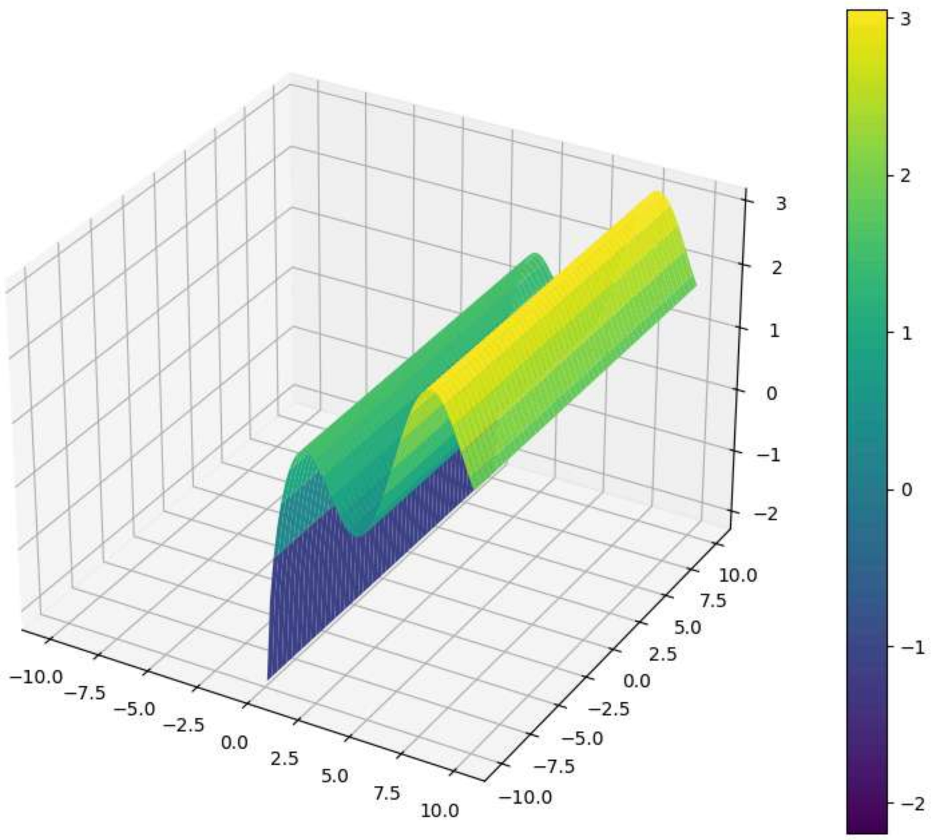

```
z = np.sin(xx) + np.cos(yy)

fig = plt.figure(figsize=(12,8))

ax = plt.subplot(projection='3d')

p = ax.plot_surface(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
<matplotlib.colorbar.Colorbar at 0x7f59eb2b7770>
```



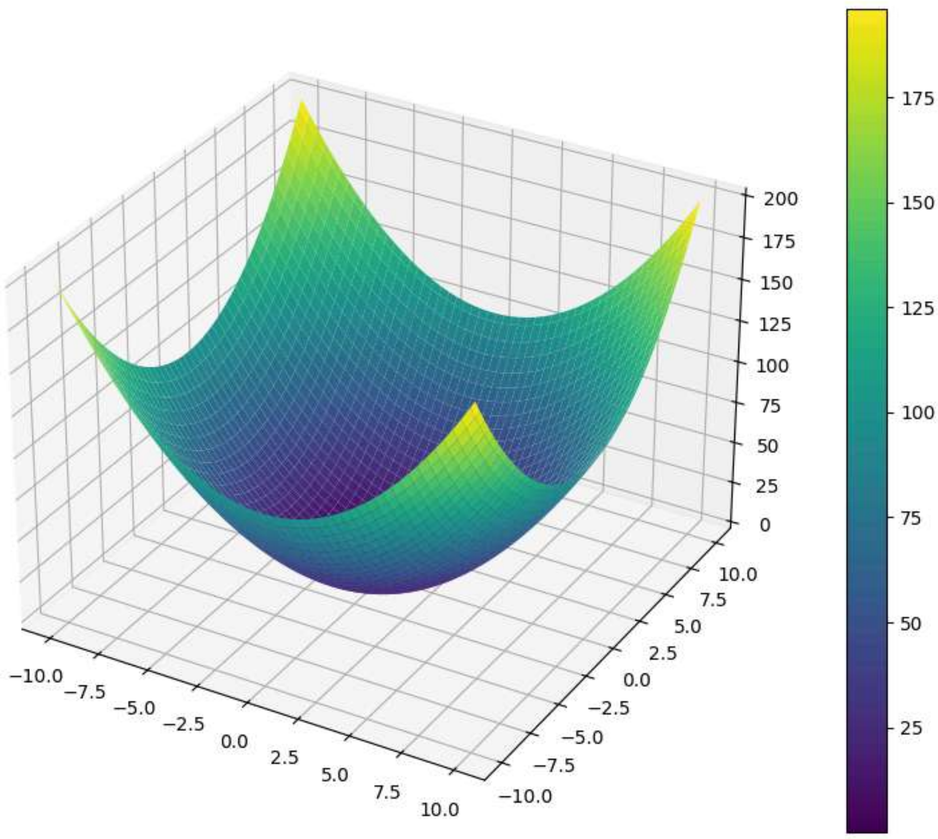

```
z = np.sin(xx) + np.log(xx)

fig = plt.figure(figsize=(12,8))

ax = plt.subplot(projection='3d')

p = ax.plot_surface(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
/tmp/ipykernel_1166/684431574.py:1: RuntimeWarning: invalid value encountered in log
  z = np.sin(xx) + np.log(xx)
<matplotlib.colorbar.Colorbar at 0x7f59e1a67a10>
```




```
fig = plt.figure(figsize=(12,8))

ax = plt.subplot(projection='3d')

p = ax.plot_surface(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

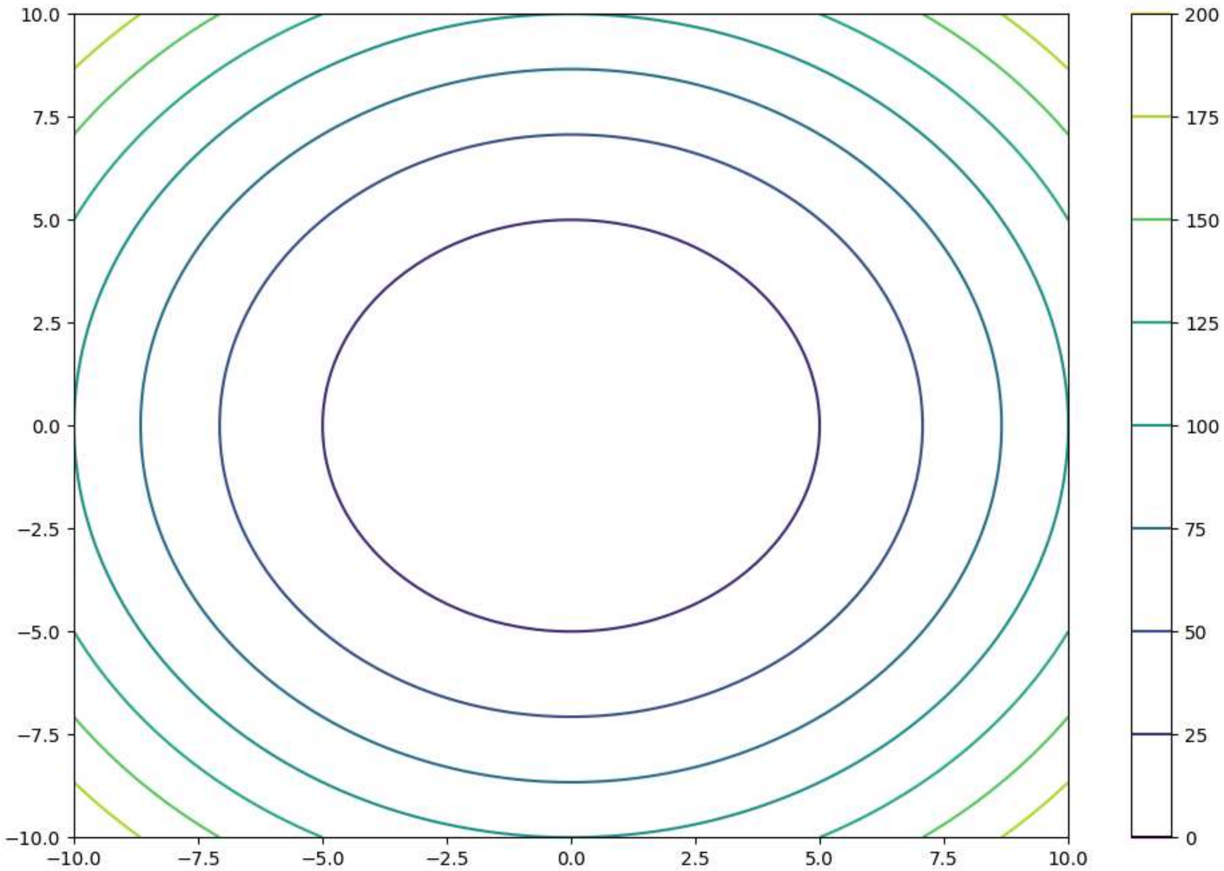
```
<matplotlib.colorbar.Colorbar at 0x7f59e1c4f770>
```



Contour Plots

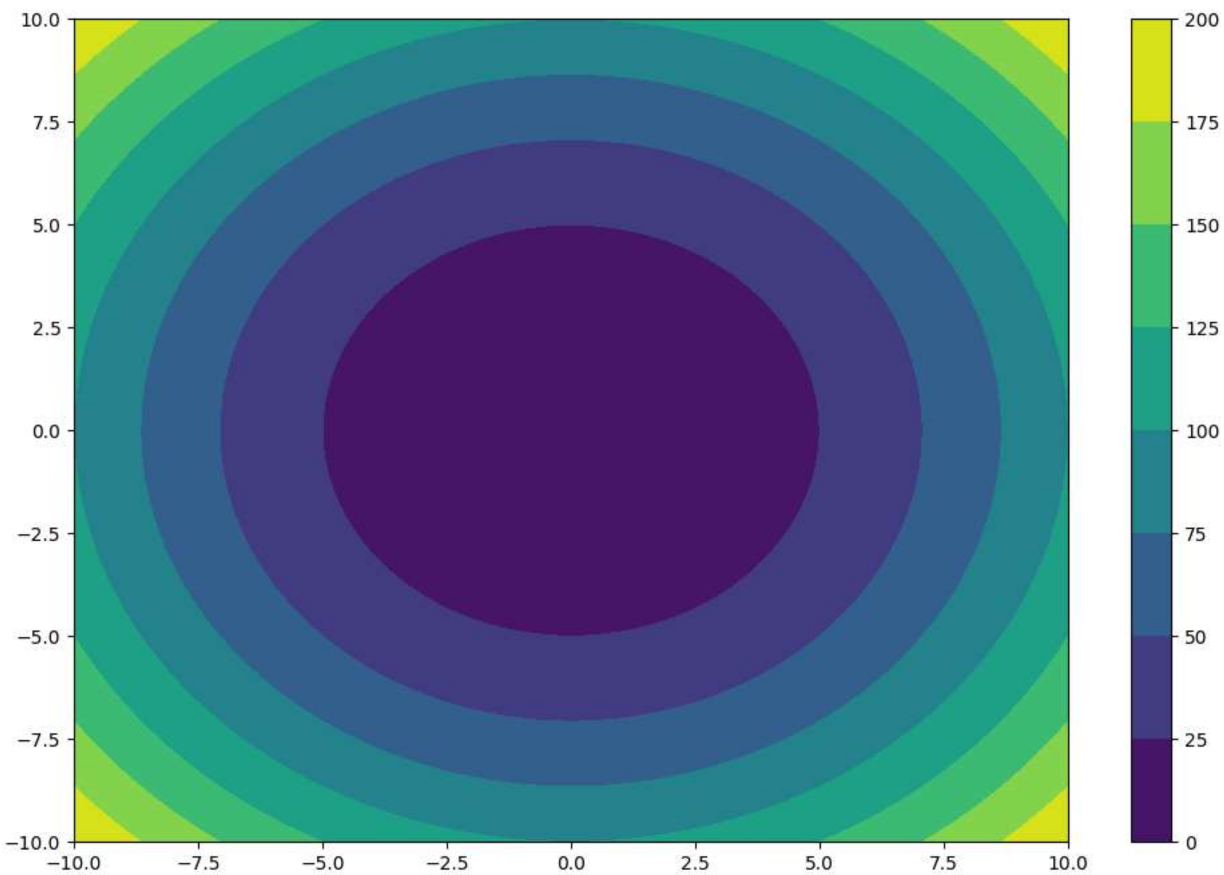
```
fig = plt.figure(figsize=(12,8))

ax = plt.subplot()

p = ax.contour(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
<matplotlib.colorbar.Colorbar at 0x7f59e15d27e0>
```



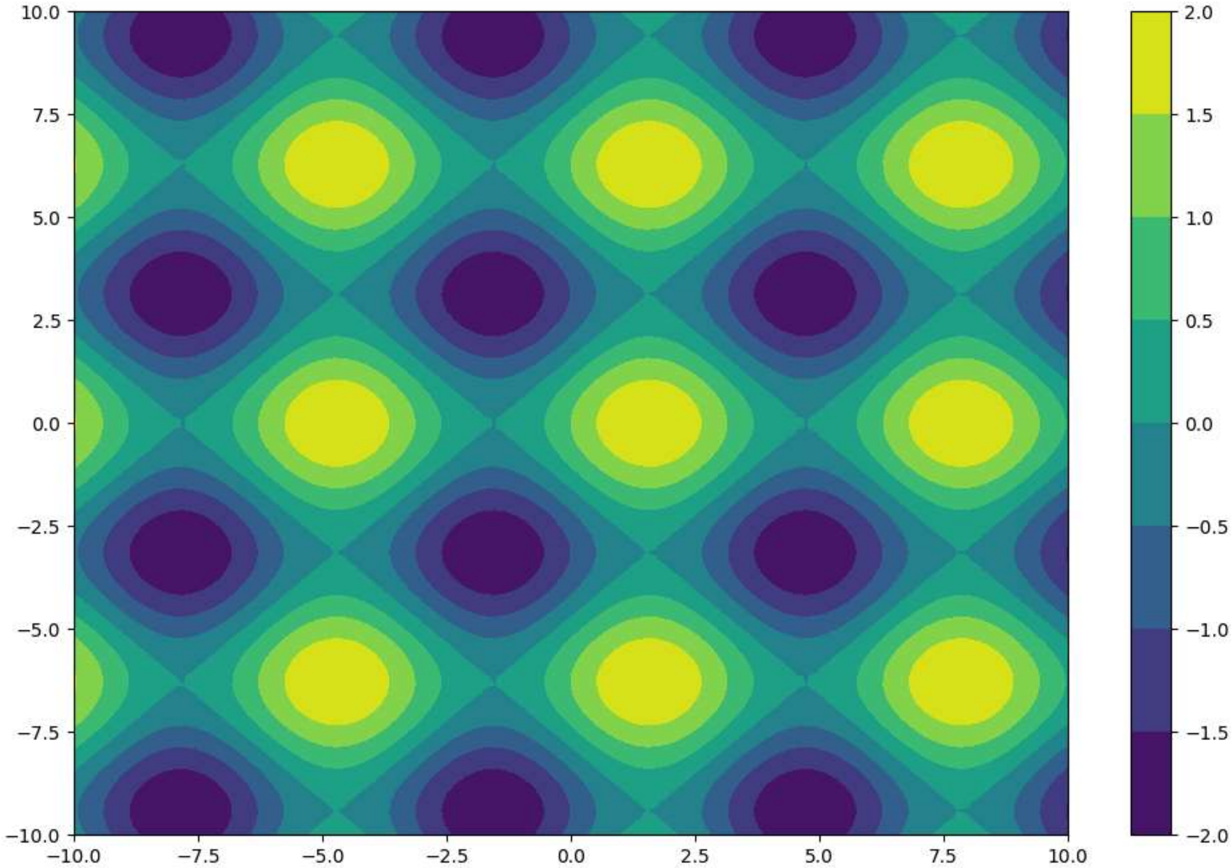

```
fig = plt.figure(figsize=(12,8))

ax = plt.subplot()

p = ax.contourf(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
<matplotlib.colorbar.Colorbar at 0x7f59e1c5dc40>
```




```
z = np.sin(xx) + np.cos(yy)

fig = plt.figure(figsize=(12,8))

ax = plt.subplot()

p = ax.contourf(xx,yy,z,cmap='viridis')
fig.colorbar(p)
```

```
<matplotlib.colorbar.Colorbar at 0x7f59e14a3710>
```



Heatmap

```
delivery = pd.read_csv('/content/IPL_Ball_by_Ball_2008_2022.csv')
delivery.head()
```

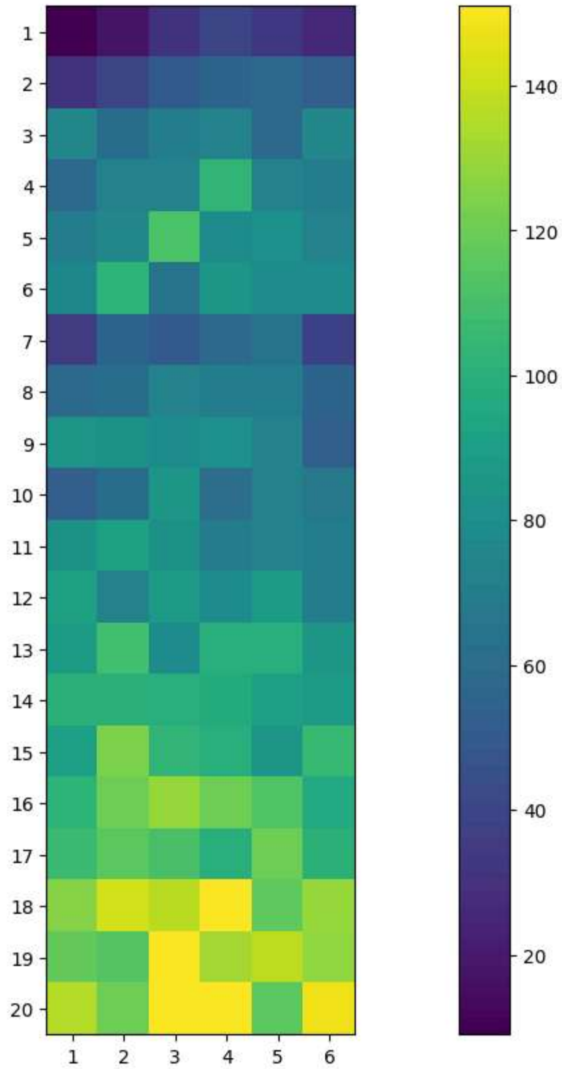
	ID	innings	overs	ballnumber	batter	bowler	non-striker	extra_type	batsman_run	extras_run	total_run	non_boundary	isWicketDelivery
0	1312200	1	0	1	YBK Jaiswal	Mohammed Shami	JC Buttler	NaN	0	0	0	0	0
1	1312200	1	0	2	YBK Jaiswal	Mohammed Shami	JC Buttler	legbyes	0	1	1	0	0
2	1312200	1	0	3	JC Buttler	Mohammed Shami	YBK Jaiswal	NaN	1	0	1	0	0
3	1312200	1	0	4	YBK Jaiswal	Mohammed Shami	JC Buttler	NaN	0	0	0	0	0
4	1312200	1	0	5	YBK Jaiswal	Mohammed Shami	JC Buttler	NaN	0	0	0	0	0

```
temp_df = delivery[(delivery['ballnumber'].isin([1,2,3,4,5,6])) & (delivery['batsman_run']==6)]
```

```
grid = temp_df.pivot_table(index='overs',columns='ballnumber',values='batsman_run',aggfunc='count')
```

```
plt.figure(figsize=(20,10))
plt.imshow(grid)
plt.yticks(delivery['overs'].unique(), list(range(1,21)))
plt.xticks(np.arange(0,6), list(range(1,7)))
plt.colorbar()
```

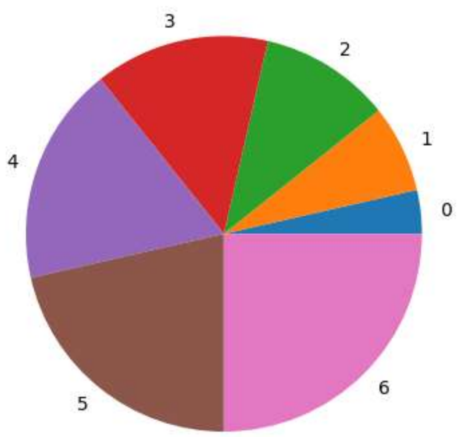
```
<matplotlib.colorbar.Colorbar at 0x7f59e1230410>
```



Pandas Plot()

```
# on a series

s = pd.Series([1,2,3,4,5,6,7])
s.plot(kind='pie')
```

```
<Axes: >
```




```
# can be used on a dataframe as well
```

```
import seaborn as sns
tips = sns.load_dataset('tips')
```

```
tips['size'] = tips['size'] * 100
```

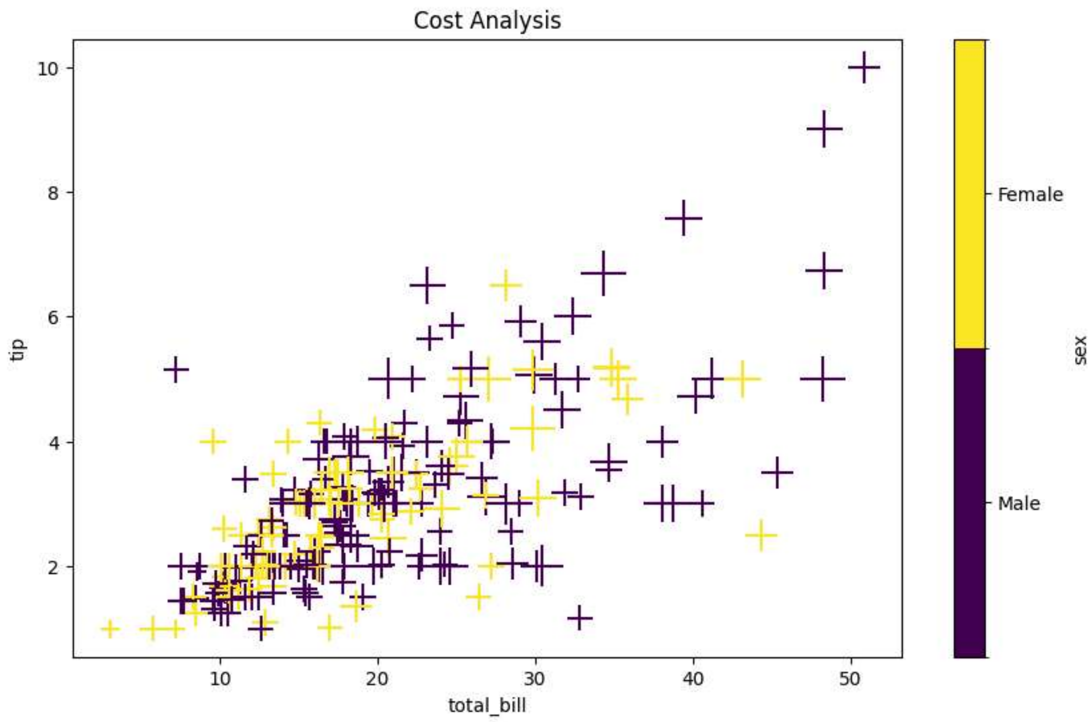
```
Start coding or generate with AI.
```

```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	200
1	10.34	1.66	Male	No	Sun	Dinner	300
2	21.01	3.50	Male	No	Sun	Dinner	300
3	23.68	3.31	Male	No	Sun	Dinner	200
4	24.59	3.61	Female	No	Sun	Dinner	400

```
# Scatter plot -> labels -> markers -> figsize -> color -> cmap
tips.plot(kind='scatter',x='total_bill',y='tip',title='Cost Analysis',marker='+',figsize=(10,6),s='size',c='sex',cmap='viridis')
```

```
<Axes: title={'center': 'Cost Analysis'}, xlabel='total_bill', ylabel='tip'>
```

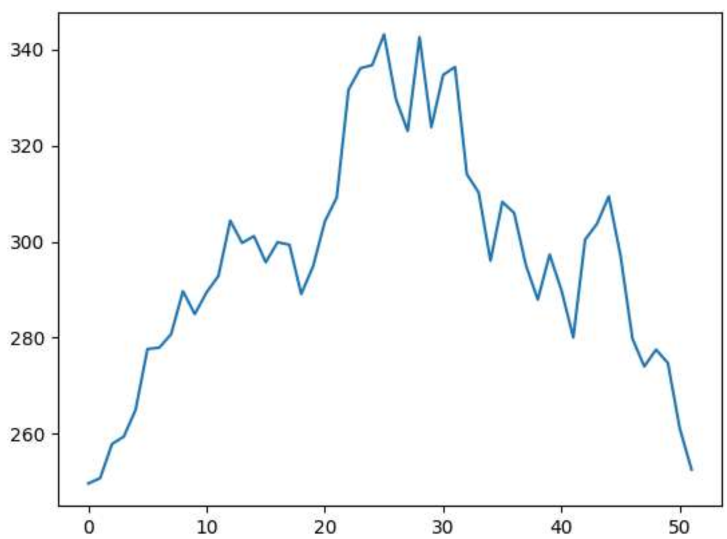



```
# 2d plot
# dataset = 'https://raw.githubusercontent.com/m-mehdi/pandas_tutorials/main/weekly_stocks.csv'

stocks = pd.read_csv('https://raw.githubusercontent.com/m-mehdi/pandas_tutorials/main/weekly_stocks.csv')
stocks.head()
```

	Date	MSFT	FB	AAPL
0	2021-05-24	249.679993	328.730011	124.610001
1	2021-05-31	250.789993	330.350006	125.889999
2	2021-06-07	257.890015	331.260010	127.349998
3	2021-06-14	259.429993	329.660004	130.460007
4	2021-06-21	265.019989	341.369995	133.110001

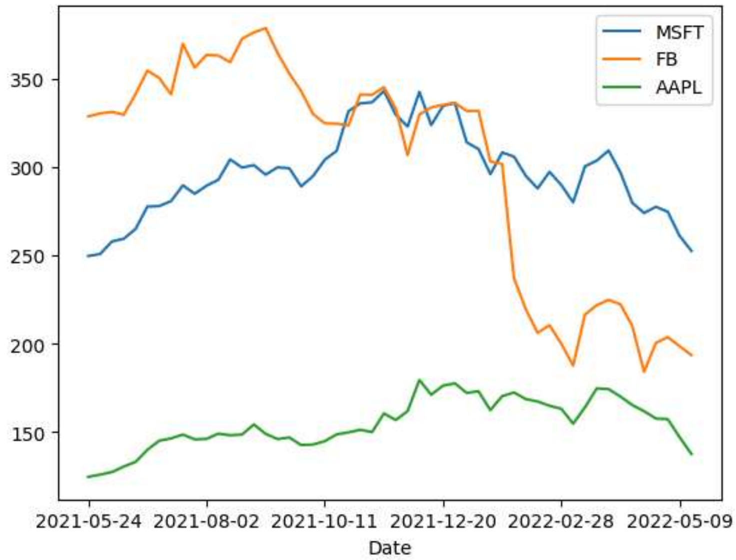
```
# line plot
stocks['MSFT'].plot(kind='line')
```

```
<Axes: >
```



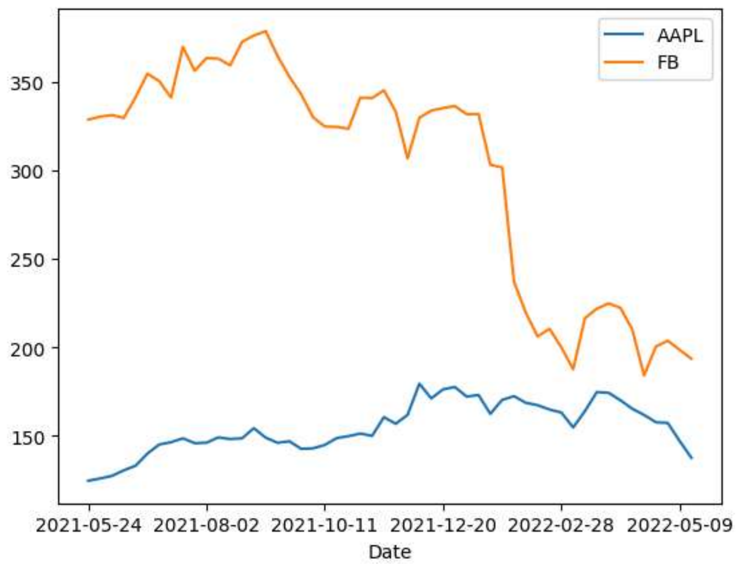

```
stocks.plot(kind='line',x='Date')
```

```
<Axes: xlabel='Date'>
```




```
stocks[['Date','AAPL','FB']].plot(kind='line',x='Date')
```

```
<Axes: xlabel='Date'>
```

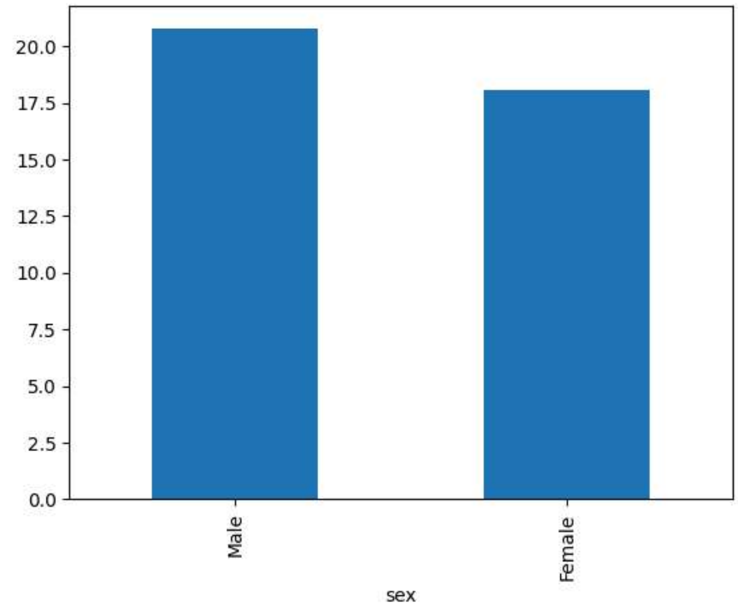



```
# bar chart -> single -> horizontal -> multiple
# using tips
temp = pd.read_csv('/content/batsman_season_record.csv')
temp.head()
```

	batsman	2015	2016	2017
0	AB de Villiers	513	687	216
1	DA Warner	562	848	641
2	MS Dhoni	372	284	290
3	RG Sharma	482	489	333
4	V Kohli	505	973	308

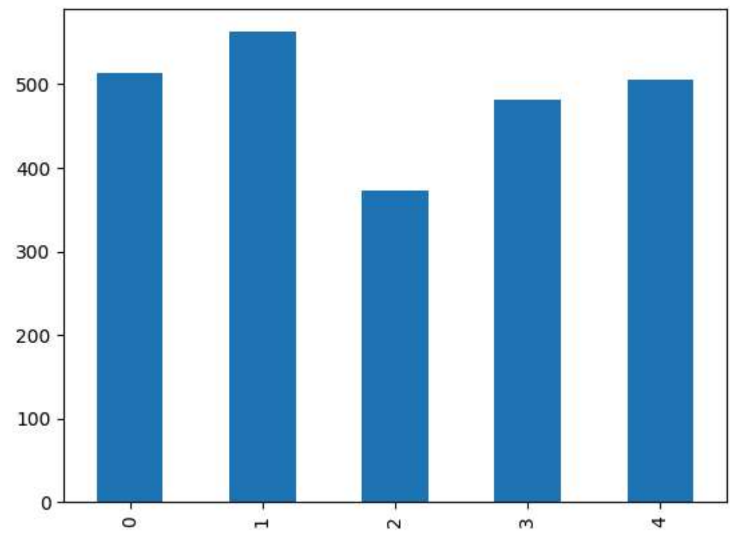
```
tips.groupby('sex')['total_bill'].mean().plot(kind='bar')
```

```
/tmp/ipykernel_1166/709106644.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of
  tips.groupby('sex')['total_bill'].mean().plot(kind='bar')
<Axes: xlabel='sex'>
```



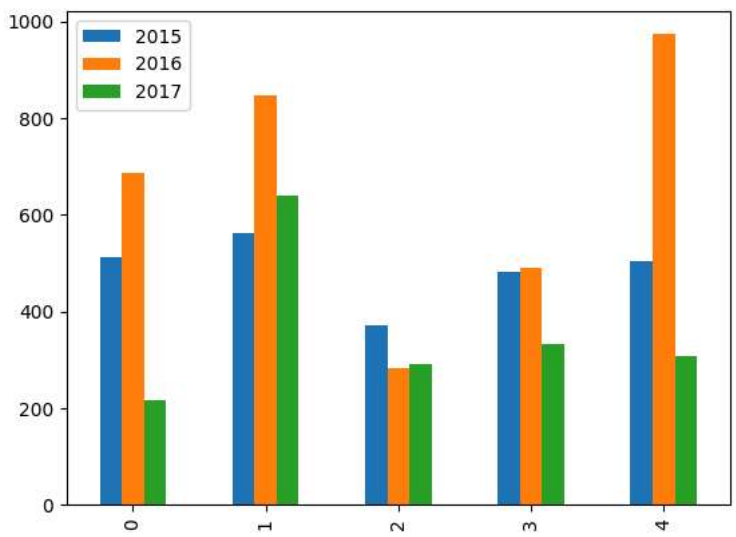

```
temp['2015'].plot(kind='bar')
```

```
<Axes: >
```



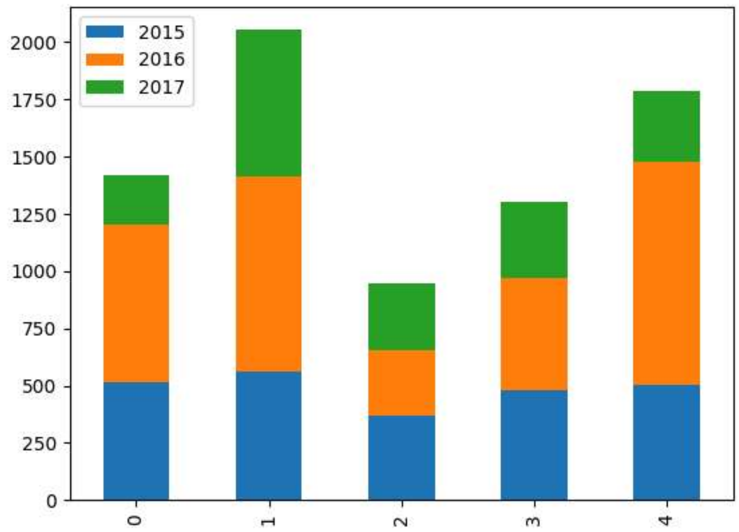

```
temp.plot(kind='bar')
```

```
<Axes: >
```



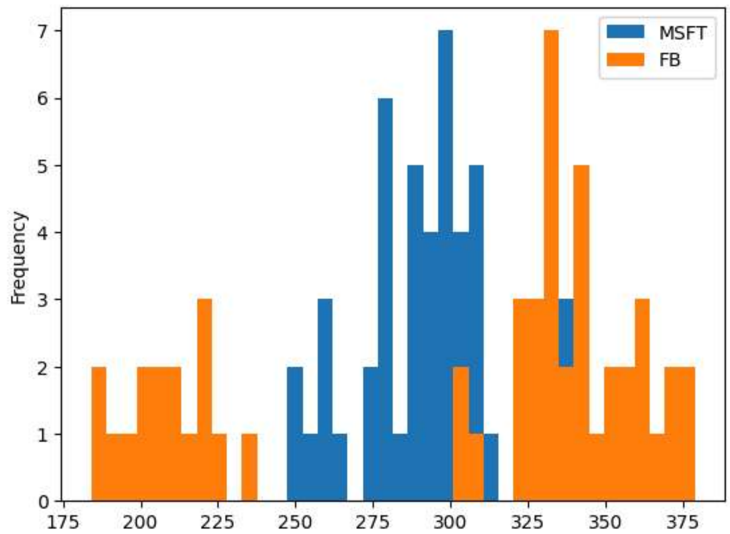

```
# stacked bar chart
temp.plot(kind='bar',stacked=True)
```

```
<Axes: >
```




```
# histogram
# using stocks

stocks[['MSFT','FB']].plot(kind='hist',bins=40)
```

```
<Axes: ylabel='Frequency'>
```

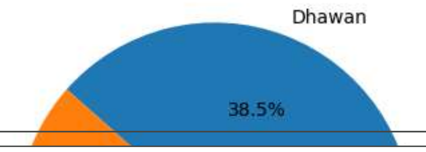



```
# pie -> single and multiple
df = pd.DataFrame(
    {
        'batsman':['Dhawan','Rohit','Kohli','SKY','Pandya','Pant'],
        'match1':[120,90,35,45,12,10],
        'match2':[0,1,123,130,34,45],
        'match3':[50,24,145,45,10,90]
    }
)

df.head()
```

	batsman	match1	match2	match3
0	Dhawan	120	0	50
1	Rohit	90	1	24
2	Kohli	35	123	145
3	SKY	45	130	45
4	Pandya	12	34	10

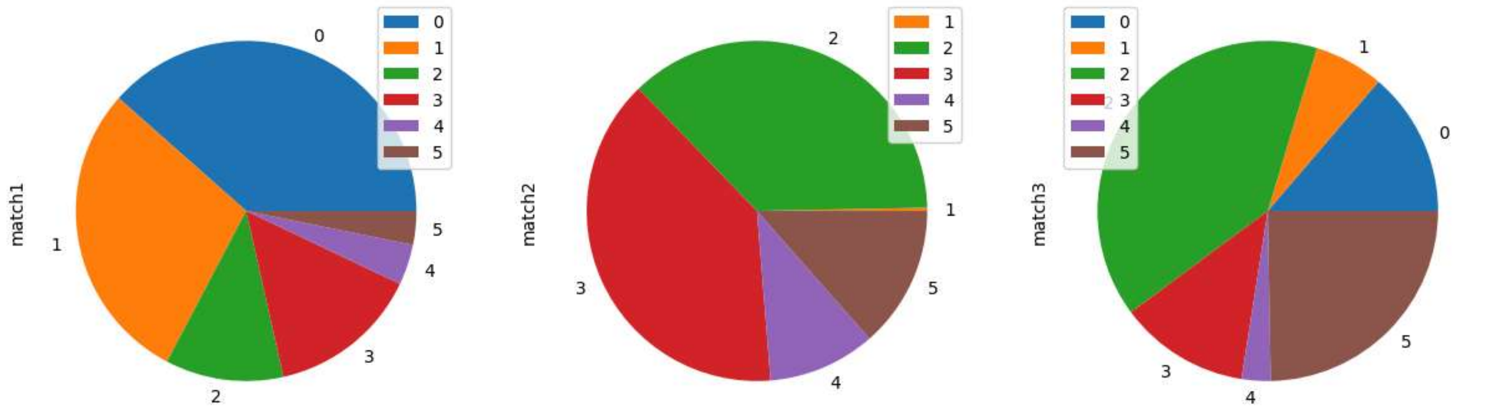
```
df['match1'].plot(kind='pie',labels=df['batsman'].values,autopct='%0.1f%%')
```

```
<Axes: ylabel='match1'>
```




```
# multiple pie charts

df[['match1','match2','match3']].plot(kind='pie',subplots=True,figsize=(15,8))
```

```
array([<Axes: ylabel='match1'>, <Axes: ylabel='match2'>,
       <Axes: ylabel='match3'>], dtype=object)
```




```
# multiple separate graphs together
# using stocks

stocks.plot(kind='line',subplots=True)
```

```
array([<Axes: >, <Axes: >, <Axes: >], dtype=object)
```